

PES2UG23CS678_TDL_Unit2

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Subject: Topics in Deep Learning (UE23CS343BB2)

Task 1:

Aim: To build a model that can automatically answer the questions from a given text passage or conversational context.

Parts completed over here:

a.) Span-based QA dataset from Wikipedia.

Output Screenshot:

The datasets used over here are as follows:

1. dev-v1.1
2. train-v1.1

b.) Conversational question answering with context-dependent answers

Output Screenshot:

The datasets used over here are as follows:

1. coqa-dev-v1.0
2. coqa-train-v1.0

c.) Chaii dataset:

Output Screenshot:

Here, we would be doing the Question-answering tasks in Hindi and Tamil. This dataset would encompass a range of topics, including literature and cultural texts.

Theory:

Here, we would be using the support of the above two datasets in order to have a Question-Answering system. Owing to the time constraints of building a new model for the same, we shall use the help of the BERT model (the pre-trained model), on which the fine-tuning would be done in order to gain the optimal answers, while keeping the execution time low. As a part of the defensive programming, we have made the checks on whether we are having an entry of the length zero or not.

Task 2:

Aim: To build a model that generates a short summary given a long article or document.

Parts completed over here:

a.) Summarization of the daily articles:

Output Screenshot:

Here, we would be using the article_highlights.csv as the dataset.

b.) summarization of the daily articles (in the local languages):

Output Screenshot:

Here, we are working with the languages of English, Hindi and Gujarati, along with the sample submission stating the expected submission to be done. So, we would be using the help of the following datasets:

1. Eng_train.csv
2. Eng_val_article.csv
3. Guj_train.csv
4. Guj_val_article.csv
5. Hin_train.csv
6. Hin_val_article.csv
7. sample_submission.csv

c.) Working with the reward models from OpenAI:

Output Screenshot:

Here, we would be getting a unique and comprehensive corpus for natural language processing tasks, with a specific emphasis on the text summarization tools for validating reward models from OpenAI. It contains columns that provide summaries of text from the TL;DR, Convolution Neural Network(abbreviated to give us the term CNN), and Daily Mail datasets, along with additional information including choices made by the workers when summarizing the text, batch information provided to differentiate different summaries created by the workers, and dataset attribute splits. So, the datasets used by us are given as follows:

1. axis_text.csv
2. axis_validation.csv
3. comparisons_train.csv
4. comparisons_validation.csv

Theory:

So, we would be building our model, which would be useful in summarization of long texts. Owing to the time constraints of building a new model for the same, we shall use the help of the BERT model (the pre-trained model), on which the fine-tuning would be done to gain the summary of the given input text (with the word limit included too), while keeping the execution time low. As a part of the defensive programming, we have made the checks on whether we are having an entry of the length zero or not.